PRIMATES, 41(2): 137–145, April 2000

Effects of Six-day Maternal Separation on Tonkean Macaque Infants

L. DRAGO and B. THIERRY
CNRS and *Université Louis Pasteur*

ABSTRACT. Maternal separation may induce a depressive state in infant macaques. However, this does not occur in all macaque species. From present hypotheses, it may be predicted that infants belonging to a species characterized by open and tolerant social relationships should not develop severe depressive symptoms. Tonkean macaques (*Macaca tonkeana*) stand as such a species. The goal of the study was to verify that the infant's reaction to mother loss is related to the social environment. The mothers of eight 5- to 9-month-old infants were removed during 6-day experimental periods. Infants' behavior was characterized by a mild initial protest stage, followed by a slight decrease in activity during later maternal separation, and quick recovery after the mother's return. No despair stage occurred. During separation, group members compensated for mother's absence by cradling the infants. That social networks determine the intensity of the infant's response to separation has far-reaching implications with regard to the meaning of depression occurrence within social networks.

Key Words: Depression; Loss; Parental; Alloparental; *Macaca tonkeana*; Sulawesi.

INTRODUCTION

Early experiments showed that mother loss may induce a depressive state in rhesus macaques (*Macaca mulatta*). Following maternal separation, some infants displayed a biphasic response characterized by an initial stage of hyperactivity accompanied by loud vocalizations, and a subsequent period of passivity featured by social withdrawal, a sharp decrease in play, and the development of a typical slouched posture (SEAY et al., 1962; HINDE et al., 1966). Such patterns mirrored the protest and despair stages reported by SPITZ (1946) and BOWLBY (1960) in human children separated from their mother. These results were replicated in several species of monkeys and apes (MINEKA & SUOMI, 1978; CODNER & NADLER, 1984). The age and sex of the infant had little effect on the intensity of the infant's response under 1 yr of age, but wide individual variation was reported (LEWIS et al., 1976; SPENCER-BOOTH & HINDE, 1971; MINEKA & SUOMI, 1978).

Important interspecific differences were also found, signs of depression being minimal or absent in some species (MINEKA & SUOMI, 1978; DOLHINOW, 1980). By comparing pigtailed (*M. nemestrina*) and bonnet macaques (*M. radiata*), KAUFMAN and ROSENBLUM (1969) showed that most of the variance in the infant's response might be explained by the contrasting social environments provided by each of these two closely related species. In bonnet macaques, mothers were permissive, they allowed their infants to contact and establish relationships with other group members. When they were separated, the infants were cradled by other females. This was believed to alleviate their distress and consequently no behavioral depression occurred. In pigtailed macaques, on the other hand, interindividual tolerance for other infants was less developed, and mothers limited their infant's interactions with other adults; consequently, no group members took care of infants during separation from their mother and they displayed severe

depressive symptoms. Other studies further stressed the role of the social context in shaping the infant's behavioral and physiological response (REITE et al., 1978, 1989). In particular, it appeared that bonnet infants might also become depressed if placed in a social environment which deprived them from substitute caregivers (KAUFMAN & STYNES, 1978; REITE & SNYDER, 1982; BOCCIA et al., 1991).

The above findings indicate that variation in infant's reaction to separation might not be primary but secondary to species differences in social relationships. While such a statement has important implications with regard to the functional meaning of depression and the mechanisms operating within societies as well, it lies on a limited number of species. Recent advances in the study of macaque social organizations have stated that they may be ordered on a scale ranging from highly hierarchical and nepotistic systems to more tolerant systems, in which the burden of power asymmetry and kinship ties is less pronounced. Many social characteristics including appeasement and socialization patterns consistently covary with such features (THIERRY, 1986, 1990, in press; DE WAAL & LUTTRELL, 1989; AURELI et al., 1997; PETIT et al., 1997). Pigtailed and rhesus macaques, where maternal separation induces a severe depressive state (KAUFMAN & ROSENBLUM, 1967; SEAY et al., 1962; SPENCER-BOOTH & HINDE, 1971; REITE et al., 1981), belong to the restrictive side of the scale. Bonnet and Tonkean macaques (*M. tonkeana*) are located toward the other end of the scale; both species are characterized by relaxed dominance relationships and nepotism, and an elevated degree of mother's permissiveness (KAUFMAN & ROSENBLUM, 1969; ROSENBLUM, 1971; CAINE & MITCHELL, 1979; THIERRY, 1985, 1986, in press; THIERRY et al., 1994). While infant bonnet macaques do not display signs of depression when left in their home group, Tonkean macaques have not been studied with regard to this phenomenon yet. The latter species thus provides an opportunity to confirm that the infant's behavior depends on the relationship network.

The present study was designed to assess the effects of a 6-day maternal separation on eight infant Tonkean macaques. During social separations, great attention was paid to the behavioral state and health of the infants. Four separate experiments were conducted: during experiment one only one mother was removed, in two other experiments two mothers were removed at a time, and in a fourth experiment three infants were simultaneously separated from their mothers. Given what is known about their social networks, two predictions were tested in Tonkean macaques: (1) group members should care for separated infants; and (2) infants should present a relatively mild response to maternal separation.

METHOD

SUBJECTS AND ENVIRONMENT

The study was carried out on eight Tonkean macaque infants aged 5 to 9 months. Information about mother-infant pairs is provided in Table 1. The subjects belonged to a social group bred at the Centre de Primatologie of the Université Louis Pasteur, Strasbourg. The group was constituted in 1978 and was composed of two matrilines. During the study periods in 1995 and 1996, it contained 26–28 individuals of various age-and-sex classes, including 11 adult females and 2–3 adult males. All subjects were captive bred except one post-reproductive female who was born in the wild.

Animals were usually free-ranging in a 1-ha wooded park. However, during each study period, the group was kept for one month in an enclosure located in the park. Animals were accustomed to this procedure, which occurred every year at spring to protect trees in bud; this

Table 1. Information about subjects.

Subjects	Age(weeks)	Sex	Mother	Number of infants separated at a time
Kis	30	Female	Eri^{p,1}	1
Vol	22	Male	Veo^{m,1}	2
Jea	20	Female	Bou^{m,1}	2
Kro	29	Male	Mar^{m,2}	2
Kal	25	Female	Elo^{p,1}	2
Jan	22	Male	Jul^{m,2}	3
Mil	30	Male	Mar^{m,2}	3
Vir	35	Male	Veo^{m,1}	3

p: Primiparous; m: multiparous; 1: first lineage; 2: second lineage.

did not induce noticeable behavioral disturbances within the group. The enclosure was constituted of an outdoor and an indoor sections of 20 m^2 each, provided with perches and shelves. Commercial monkey pellets and water were available ad libitum in the indoor part.

Experimental Procedure

The procedure involving the removal of the mother took place four times in different study periods. There could be one, two, or three mothers removed at time (Table 1). On day 0 at 14:00, mothers were anesthetized (Ketamine hydrochloride, I.M., 8 mg/kg) and removed from the group. They were then isolated from the group during six days. Mothers were reintroduced on day 6; they were anesthetized at 11:30, they entered the group at 14:00, after awakening.

During observations, the group was locked in the outdoor part of the enclosure. Two different observation schedules were followed. The first was designed to assess the short-term response of infants to separation and reunion. The subjects were observed from 15:30 to 17:30 on days –3 and +9 (control days) and days 0 and 6 (separation and reunion days).

A second observation schedule was designed to assess the prolonged effects of separation. It occurred in the morning from 09:00 to 12:00. One half-hour before the start of observations, the group received apples and carrots; some pieces of fruits and vegetables remained on the ground until midday. The study period lasted 18 days, during which the subjects were observed 2 mornings out of 3 days. This resulted in the subjects being observed four mornings before separation (pre-separation phase: days –5 to 0), during separation (separation phase: days +1 to +6), and after reunion (post-separation phase: days +7 to +12).

Behavior Measurements

Subjects were observed 10 min every half an hour, thus a total per subject of 60 min per morning or 40 min per afternoon. Occurrences of partner active contacts and leaves were recorded using animal focal sampling (Altmann, 1974). Screeching (squeaks, geckers, or screams) and coo vocalizations were sampled with the same method; calls separated by intervals of at least 3-sec were counted as distinct utterances. During the same 10-min sampling periods, activities and contacts were scored every minute using instantaneous sampling (Altmann, 1974). The following activities were recorded: clinging (ventro-ventral contact with a partner while awake or asleep), feeding (picking or ingesting food), exploration (locomotion, solitary play, or environmental manipulation), looking out (awake and inactive, alone or in contact but not clinging on a partner), self-directed behaviors (grooming, licking, clasping, or manipulation of own's body), social grooming, social play (chase or wrestle), social interaction

(excluding social play and grooming). Every minute, the observer also recorded whether the subject was in contact with a partner. Lastly, the observer was prepared to record any occurrence of the slouch behavior characteristic of the depressive state: sitting inactive in a hunched-over posture.

DATA ANALYSIS

For each behavior recorded in instantaneous sampling, numbers of scans were processed as percentages of the total number of scans. Vocalizations recorded using focal animal sampling were processed as hour frequencies. Active contact and leave occurrences were used to calculate a contact responsibility index defined as the number of contacts by the subject plus the number of leaves by the partner divided by the number of leaves by the infants plus the number of contacts by the partner (note that contacts and leaves occurring in the play context were not included in the computation of the index). Categories of partners were defined as follows: mother, adult female (≥ 5 yr old), adolescent female (2–4 yr old), adult male (≥ 6 yr old), adolescent male (2–5 yr old), juvenile (< 2 yr old). Kinship was defined by maternal lineage membership.

Data were analyzed using MANOVAs (Superanova software, Abacus Concepts Inc.). As individuals differed in many parameters (Table 1), no attempt was made to compare them. With regard to the data recorded in the morning, data were averaged as 2-day means; every observation phase was therefore composed of two blocks, each one distributed over three days. These data were analyzed using a nested MANOVA with blocks and phases as variables. Variances could differ among days or 2-day blocks since number of subjects and available partners varied according to experiments. Thus, data were transformed in their arcsine values for percentages and square root values for other measures. The homogeneity of variances was then checked using the Bartlett's test (SOKAL & ROHLF, 1981). Variances were found to differ for only one variable (percentage of scans in contact with adolescent females), which was consequently analyzed using the Friedman's non-parametric ANOVA (SIEGEL & CASTELLAN, 1988). Significance level was set at 0.05.

RESULTS

INITIAL EFFECTS OF SEPARATION AND REUNION

Comparison of the four sets of data recorded during the afternoon showed that several of the infants' behaviors were affected by separation and reunion events. Percentages of scans in clinging were more elevated in the hours following reunion with the mother (Table 2). In contrast to control days, looking out was higher after separation and lower after reunion (Table 2). A decrease in social play occurred after separation or reunion (Table 2). The frequencies of coos and screechings increased significantly following separation. Rates of self-directed behaviors were similarly heightened albeit at a lesser degree (Table 2). No statistically significant fluctuations appeared in the rates of exploration, social grooming, and social interactions (Table 2).

PROLONGED EFFECTS OF SEPARATION AND REUNION

The prolonged effects of separation and reunion on the infants' behaviors were assessed by comparing the three phases and six blocks of data recorded in the morning. No statistically significant block effects or block-phase interactions were found.

Table 2. Initial effects of separation and reunion.

Variables	Earlier control day	Separation day	Reunion day	Later control day	Day effect $F_{3,28}$	p
Clinging[1)]	43.0 ± 11.7	18.4 ± 9.6	76.9 ± 6.7	33.4 ±10.1	6.3	.002
Exploration[1)]	27.8 ± 7.6	35.0 ± 8.0	15.0 ± 7.2	27.8 ± 7.0	2.0	.143
Looking out[1)]	9.7 ± 3.1	30.0 ± 8.5	4.4 ± 1.5	12.5 ± 3.8	5.2	.006
Self-directed[1)]	0	3.6 ± 1.1	0	0.6 ± 0.4	14.1	.0001
Social interaction[1)]	2.8 ± 1.0	4.4 ± 1.8	3.1 ± 1.8	5.3 ± 1.9	0.4	.738
Social grooming[1)]	0.6 ± 0.6	3.4 ± 2.4	0.6 ± 0.4	3.1 ± 3.1	0.7	.558
Social play[1)]	16.9 ± 3.8	4.4 ± 2.0	1.3 ± 0.9	12.2 ± 4.9	5.2	.005
Coo[2)]	2.8 ± 2.6	23.1 ± 6.4	3.2 ± 2.8	4.7 ± 4.7	7.8	.001
Screeching[2)]	0.8 ± 0.6	10.9 ± 4.4	2.3 ± 1.8	0	6.4	.002

Values represent means ± SE. Statistical analyses were carried out on transformed data (arcsine values for percentages, square root values for frequencies). 1) Percentage of scans; 2) hour frequency.

Activities and Vocalizations

Percentages of scans decreased for clinging awake during the prolonged separation phase (Table 3). Social play also decreased during separation, it returned to baseline at the end of the post-separation phase (Table 3). Feeding, looking out and distress calls (screeching, in particular) were heightened during separation (Table 3). Variations in the time spent in exploration remained statistically non significant (Table 3). No significant trends appeared for the rates of clinging asleep, social grooming, social interactions, and self-directed behaviors.

Contacts

Maternal separation and reunion did not significantly affect time spent off partners (non contact, Table 3). Infants increased their contact rates with most categories of partners during separation. The increase was especially high for adolescent females (Friedman test, Table 3). Variations in contact rates with the mother did not reach statistical significance (Table 3). Kinship did not affect contact variation patterns (Table 3).

The contact responsibility index did not significantly vary for mothers and adult females (Table 3). In contrast, infants tended to be more responsible for contact with adolescent females during separation in comparison with control phases (Table 3). It is worth adding that no instance of competition was recorded among infants for access to preferred partners.

Infant clinging to an individual other than the mother arose dramatically during mother's absence (Table 3). Cradling partners belonged to three age-and-sex classes: adolescent females (separation phase: 55.0% of cling instances; control phases: 48.9%), adult females (35.5 and 51.1%), adolescent males (9.5 and 0%). For each subject, there were between one and six possible cradling partners. While six of the infants had one major partner who represented between two thirds and 100% of cradlings, two of them had two major partners responsible for 30 to 55% of cradlings. When considering maternal kin ties between infants and partners, it appeared that they were unrelated (3 cases), sibling (3), or belonged to the same matriline (coefficient of genetic relatedness by the mother: 1 case: 0.25; 2: 0.125; 1: 0.063).

Note

With regard to the occurrence of slouch behavior, one subject (*Vol*) displayed it once during a full 10-min sampling period in a morning of the first block following mother removal. No other instances of this hunched-over posture was recorded, neither in this subject, nor in the others.

Table 3. Prolonged effects of separation and reunion.

Variables	Pre-separation phase		Separation phase		Post-separation phase		Phase effect[1]	
	1st block	2nd block	1st block	2nd block	1st block	2nd block	F (or χ^2)[2]	p
Activities[3]								
Clinging awake	36.9 ± 4.1	37.9 ± 7.7	20.0 ± 4.1	26.3 ± 8.7	34.2 ± 4.3	28.0 ± 7.6	3.2	.052
Clinging asleep	16.1 ± 4.2	21.3 ± 6.2	17.0 ± 4.2	18.1 ± 4.1	31.0 ± 4.2	17.7 ± 3.0	1.5	.229
Feeding	8.3 ± 3.3	9.2 ± 2.3	22.7 ± 4.5	22.8 ± 4.0	12.0 ± 3.7	13.2 ± 3.2	8.8	.001
Exploration	22.7 ± 3.7	17.9 ± 4.8	14.4 ± 2.9	13.7 ± 5.1	10.6 ± 2.9	16.7 ± 3.3	1.7	.203
Looking out	6.5 ± 2.3	6.1 ± 2.2	19.0 ± 6.0	14.1 ± 4.7	7.5 ± 1.7	11.2 ± 2.6	4.3	.020
Self-directed	0.6 ± 0.3	0.5 ± 0.2	0.9 ± 0.4	1.1 ± 0.7	0.8 ± 0.4	0.8 ± 0.4	0.2	.828
Social interaction	2.2 ± 0.9	2.9 ± 1.0	3.2 ± 1.4	1.8 ± 0.6	3.0 ± 1.3	4.1 ± 1.0	0.4	.649
Social grooming	4.8 ± 1.3	2.4 ± 1.3	6.0 ± 2.2	4.3 ± 1.4	4.7 ± 1.1	2.6 ± 1.2	0.3	.732
Social play	7.9 ± 1.4	9.5 ± 3.6	3.3 ± 1.1	3.8 ± 1.7	2.9 ± 0.7	10.2 ± 2.3	3.8	.030
Contacts[3]								
Mother	60.4 ± 5.7	63.3 ± 8.3			70.5 ± 4.5	52.0 ± 6.6	0.09	.764
Adult females	5.1 ± 1.6	5.5 ± 1.6	14.3 ± 5.4	9.9 ± 1.7	3.4 ± 0.7	5.5 ± 1.0	7.1	.002
Adolescent females	3.1 ± 1.6	2.2 ± 1.0	31.5 ± 6.6	42.7 ± 13.3	5.4 ± 1.5	5.0 ± 1.3	16.7	.004
Adult males	0.1 ± 0.1	0.1 ± 0.1	1.7 ± 0.9	1.7 ± 1.2	0	0.6 ± 0.2	3.9	.028
Adolescent males	3.7 ± 1.4	1.9 ± 0.7	2.8 ± 0.9	3.2 ± 1.1	1.7 ± 0.5	3.2 ± 0.6	0.04	.966
Juveniles	5.1 ± 1.7	5.5 ± 1.6	14.3 ± 5.4	9.9 ± 1.7	3.4 ± 0.7	5.5 ± 1.0	7.1	.002
Kins	66.8 ±13.1	61.4 ±11.1	65.6 ±11.8	76.8 ± 9.6	61.4 ±11.1	70.2 ± 8.5	0.1	.865
No contact	25.7 ± 3.6	25.9 ± 5.9	36.0 ± 5.7	30.6 ± 7.9	20.7 ± 3.5	32.4 ± 5.0	0.9	.420
Infant clinging to a non-mother partner	0.7 ± 0.5	1.4 ± 0.9	33.9 ± 7.2	42.9 ± 12.6	1.5 ± 0.7	1.7 ± 0.4	23.2	.0001
Contact responsibility[4]								
Mother	1.1 ± 0.1	1.3 ± 0.4			1.8 ± 0.5	1.6 ± 0.3	1.8	.185
Adult females	1.3 ± 0.6	0.9 ± 0.04	1.2 ± 0.1	1.4 ± 0.2	0.9 ± 0.1	0.9 ± 0.04	2.3	.111
Adolescent females	0.7 ± 0.1	0.6 ± 0.1	1.1 ± 0.1	1.2 ± 0.3	0.6 ± 0.1	0.7 ± 0.1	4.0	.026
Vocalizations[5]								
Coo	0.9 ± 0.4	1.7 ± 1.5	9.4 ± 4.3	7.8 ± 5.4	3.1 ± 1.6	1.8 ± 1.4	2.8	.073
Screeching	0.1 ± 0.1	0.2 ± 0.1	2.1 ± 0.6	1.9 ± 0.8	2.0 ± 0.7	0.8 ± 0.5	6.9	.003

Values represent means ± SE. 1) Statistical analyses were carried out on transformed data (arcsine values for percentages, square root values for other measures); 2) *F*, *d.f.*=2,42, except for values related to the mother for which *d.f.*=1,28, and for contacts with adolescent females for which a Friedman test yielded a χ^2, *d.f.*=5; 3) percentage of scans; 4) index; 5) hour frequency.

Although *Vol* was the son of the dominant female of the group, he had at first difficulties to find partners allowing him to cling, and showed by far the lowest score of percentage of scans in cling (9.8%) during the first block of the separation phase. In fact, he was the only subject who was never recorded in contact with an adolescent female at the time of separation. During the second block, he was cradled at normal rates (51.7% of scans) by two major partners who were adolescent males from his own matriline.

DISCUSSION

In the hours following maternal separation, activity as measured by exploration and social interactions was hardly affected. Rates of looking out, self-directed behaviors and distress calls in infants increased while social play decreased. When considering the magnitude of these changes, they appeared relatively weak. For instance, frequencies of distress calls were ten-fold higher than baseline. In similar circumstances, vocalization rates were multiplied by 50 to 100 in pigtailed and rhesus macaques, though it is difficult to assess whether measurements were comparable between studies (SPENCER-BOOTH & HINDE, 1971; SINGH, 1975; ROSENBLUM & PLIMPTON, 1981). The low intensity level of the protest stage observed in Tonkean infants matched those reported in bonnet macaques.

A mild delayed behavioral response to separation also characterized Tonkean infants. During the separation phase, rates of social interactions, and self-directed behaviors did not change while looking out, food intake, and frequencies of distress calls were heightened. Decreases in exploration –including locomotion– were not significant and the subjects still played, albeit at a lower rate. While such results are consistent with those found in bonnet macaques, they are at odds with results reported in other macaque species (KAUFMAN & ROSENBLUM, 1967, 1969; SPENCER-BOOTH & HINDE, 1971; REITE et al., 1981, 1989). In particular, no extended periods of inactivity in a hunched posture were observed. A single bout of slouch behavior was recorded in one of the infants, and no strong reduction in interactions and movement patterns took place. It may be asserted that no despair stage occurred in the subjects. As in bonnet macaques, and contrary to rhesus and pigtailed macaques, a 6-day maternal separation did not induce severe behavioral disturbance in Tonkean macaques.

Mild response to separation was associated with quick recovery after the mother's return. Rates of infant clinging on mother were not significantly heightened in the days following reunion (cf. KAUFMAN & ROSENBLUM, 1969; SPENCER-BOOTH & HINDE, 1971; REITE et al., 1981, 1989). Behavioral parameters went back to baseline in the days following reunion. In contrast, rhesus infants needed some weeks to return to pre-separation levels after a 6-day maternal separation (SPENCER-BOOTH & HINDE, 1971).

The minimal disruptive consequences of mother loss in Tonkean infants may be accounted for by the behavior of group members. Infants were allowed to cling and sleep on partners during separation at levels similar to those reported for bonnet macaques (KAUFMAN & ROSENBLUM, 1969; REITE et al., 1989). Their contact rates were consistently high. Contact occurred with many group members but adolescent females appeared as the most available partners and also the most sought-after by the infants during separation. Each infant had one or two major caregiving figures responsible for most cradling instances (cf. REITE et al., 1989). Many of them were maternally related to the infant but some had no such kin ties.

It may be noticed that the size of the study group was two or three fold larger than the social groups investigated by previous authors. Infants might have had more opportunities receiving care from other group members. However, in the present experiments, two or three mothers could be removed simultaneously. The number of potential substitute caregivers was therefore comparable between studies. It may be safe to conclude that a group of Tonkean macaques containing 25–30 individuals can buffer the consequences of a disruption in the maternal bond for two or three infants at time.

An infant's chances of survival are influenced by the structure of the group in which it lives. In species characterized by strict dominance and nepotistic relations (e.g. rhesus and pigtailed macaques), the mother is compelled to frequently retrieve her infant, which limits the network of social relationships available to the infant. As an outcome, an exclusive mother-infant bond develops, leaving the infant helpless in the case of mother loss. Under experimental conditions, a proportion of rhesus and pigtailed infant macaques collapse or die after separation or even following reunion (SPENCER-BOOTH & HINDE, 1971; KAUFMAN & STYNES, 1978; ROSENBLUM & PLIMPTON, 1981). In the wild, losing the main source of nourishment and protection would place the infant in immediate jeopardy (DOLHINOW, 1980; ROSENBLUM & PLIMPTON, 1981). In species experiencing more open relationships (e.g. Tonkean and bonnet macaques), mother's permissiveness favors social experience in her infant. Conspecifics may provide care, warmth, and transportation in her absence, alleviating psychological disturbance and paving the way to permanent adoption (DOLHINOW, 1980; THIERRY & ANDERSON, 1986).

A commonly held view in sociobiology is that any characters of social organizations have been shaped by the action of selective pressures. However, some characters may arise as non-adaptive consequences of others (GOULD & LEWONTIN, 1979; THIERRY, 1990, 1997). The ability

of the infant to deal with mother loss remains as such a case. At the individual level, certain temperament and reactivity levels are likely adaptive in the normal situation. However, the profound depressive stage occurring in non-human primates possibly results from coping mechanisms going out of control (REITE et al., 1978; ROSENBLUM & PLIMPTON, 1981). At the collective level, the occurrence or non-occurrence of this stage may be considered a contingent phenomenon, i.e. a by-product of the species social organization. Although this may incur drastic effects in terms of infant survival, natural selection cannot act on such patterns without altering individual temperament and the whole social organization (THIERRY, 1997, in press).

Acknowledgments. We thank P. UHLRICH for his technical assistance and A. DUCOING for language advice. Regarding the ethics question, the research has been conducted with permission No. 448 to the second author (Agricultural Department: manipulation of non-human primate social groups).

REFERENCES

ALTMANN, J. 1974. Observational study of behaviour: sampling methods. *Behaviour*, 49: 227–265.

AURELI, F.; DAS, M.; VEENEMA, H. C. 1997. Differential kinship effect on reconciliation in three species of macaques (*Macaca fascicularis*, *M. fuscata*, and *M. sylvanus*). *J. Comp. Psychol.*, 111: 91–99.

BOCCIA, M. L.; REITE, M.; LAUDENSLAGER, M. L. 1991. Early social environment may alter the development of attachment and social support: two case reports. *Inf. Behav. Develop.*, 14: 253–260.

BOWLBY, J. 1960. Grief and mourning in infancy and early childhood. *Psychoanal. Stud. Child*, 15: 9–52.

CAINE, N. G.; MITCHELL, G. 1979. The relationship between maternal rank and companion choice in immature macaques (*Macaca mulatta* and *M. radiata*). *Primates*, 20: 583–590.

CODNER, M. A.; NADLER, R. D. 1984. Mother-infant separation and reunion in the great apes. *Primates*, 25: 204–217.

DE WAAL, F. B. M.; LUTTRELL, L. M. 1989. Toward a comparative socioecology of the genus *Macaca*: different dominance styles in rhesus and stumptail monkeys. *Amer. J. Primatol.*, 19: 83–109.

DOLHINOW, P. 1980. An experimental study of mother loss in the Indian langur monkey (*Presbytis entellus*). *Folia Primatol.*, 33: 77–128.

GOULD, S. J.; LEWONTIN, R. C. 1979. The spandrels of San Marco and the Panglossian paradigm: a critique of the adaptationist programme. *Proc. Roy. Soc. Lond.*, B 205: 581–598.

HINDE, R. A.; SPENCER-BOOTH, Y.; BRUCE, M. 1966. Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature*, 210: 1021–1033.

KAUFMAN, I. C.; ROSENBLUM, L. A. 1967. The reaction to separation in infant monkeys: anaclytic depression and conservation withdrawal. *Psychosom. Med.*, 29: 648–675.

KAUFMAN, I. C.; ROSENBLUM, L. A. 1969. Effects of separation from mother on the emotional behavior of infant monkeys. *Ann. N. Y. Acad. Sci.*, 159: 681–695.

KAUFMAN, I. C.; STYNES, A. J. 1978. Depression can be induced in a bonnet macaque infant. *Psychosom. Med.*, 40: 71–75.

LEWIS, J. K.; MCKINNEY, W. T.; YOUNG, L. D.; KRAEMER, G. W. 1976. Mother-infant separation in rhesus monkeys as a model of human depression: a reconsideration. *Arc. Gen. Psychiat.*, 33: 699–705.

MINEKA, S.; SUOMI, S. J. 1978. Social separation in monkeys. *Psychol. Bull.*, 85: 1376–1400.

PETIT, O.; ABEGG, C.; THIERRY, B. 1997. A comparative study of aggression and conciliation in three cercopithecine monkeys (*Macaca fuscata, Macaca nigra, Papio papio*). *Behaviour*, 134: 415–432.

REITE, M.; KAEMINGK, K.; BOCCIA, M. L. 1989. Maternal separation in bonnet monkey infants: altered attachment and social support. *Child Develop.*, 60: 473–480.

REITE, M.; SHORT, R.; KAUFMAN, I. C.; STYNES, A. J.; PAULEY, J. D. 1978. Heart rate and body temperature in separated monkey infants. *Biol. Psychiat.*, 13: 91–105.

REITE, M.; SHORT, R.; SEILER, C.; PAULEY, J. C. 1981. Attachment, loss, and depression. *J. Child Psychol. Psychiat.*, 22: 141–169.

REITE, M.; SNYDER, D. 1982. Physiology of maternal separation in a bonnet macaque infant. *Amer. J. Primatol.*, 2: 115–120.

ROSENBLUM, L. A. 1971. Kinship interaction patterns in pigtail and bonnet macaques. In: *Proceedings of the Third International Congress of Primatology, Vol. 3*, KUMMER, H. (ed.), Karger, Basel, pp. 79–84.

ROSENBLUM, L. A.; PLIMPTON, E. H. 1981. The infant's effort to cope with separation. In: *The Uncommon Child*, LEWIS, L.; ROSENBLUM, L. A. (eds.), Plenum Press, New York, pp. 225–257.

SEAY, B. M.; HANSEN, E. W.; HARLOW, H. F. 1962. Maternal separation in the rhesus monkey. *J. Child Psychol. Psychiat.*, 3: 123–132.

SIEGEL, S.; CASTELLAN, N. J. 1988. *Nonparametric Statistics for the Behavioral Sciences*. McGraw-Hill, Singapore.

SINGH, M. 1975. Mother-infant separation in rhesus monkey living in natural environment. *Primates*, 16: 471–476.

SOKAL, R. R.; ROHLF, F. J. 1981. *Biometry*. Freeman, San Francisco.

SPENCER-BOOTH, Y.; HINDE, R. A. 1971. Effects of 6-days separation from mother on 18- to 32-week-old rhesus monkeys. *Anim. Behav.*, 19: 174–191.

SPITZ, R. A. 1946. Anaclitic depression. *Psychoanal. Stud. Child*, 2: 313–342.

THIERRY, B. 1985. Social development in three species of macaque (*Macaca mulatta, M. fascicularis, M. tonkeana*): a preliminary report on the first ten weeks of life. *Behav. Proc.*, 11: 89–95.

THIERRY, B. 1986. A comparative study of aggression and response to aggression in three species of macaque. In: *Primate Ontogeny, Cognition and Social Behaviour*, ELSE, J. G.; LEE, P. C. (eds.), Cambridge Univ. Press, Cambridge, pp. 307–313.

THIERRY, B. 1990. Feedback loop between kinship and dominance: the macaque model. *J. Theor. Biol.*, 145: 511–521.

THIERRY, B. 1997. Adaptation and self-organization in primate societies. *Diogenes*, 180: 39–71.

THIERRY, B. in press. Covariation of conflict management patterns across macaque species. In: *Natural Conflict Resolution*, AURELI, F.; DE WAAL, F. B. M. (eds.), Univ. of California Press, Berkeley.

THIERRY, B.; ANDERSON, J. R. 1986. Adoption in anthropoid primates. *Int. J. Primatol.*, 7: 191–216.

THIERRY, B.; ANDERSON, J. R.; DEMARIA, C.; DESPORTES, C.; PETIT, O. 1994. Tonkean macaque behaviour from the perspective of the evolution of Sulawesi macaques. In: *Current Primatology, Vol. 2*, ROEDER, J. J.; THIERRY, B.; ANDERSON, J. R.; HERRENSCHMIDT, N. (eds.), Univ. Louis Pasteur, Strasbourg, pp. 103–117.

——Received: *February 22, 1999*; Accepted: *December 1, 1999*

Authors' Names and Address: L. DRAGO and B. THIERRY, *Laboratoire d'Ethologie des Primates, CNRS (FRE 2130), Université Louis Pasteur, 7 rue de l'Université, 67000 Strasbourg, France*. e-mail: thierry@neurochem.u-strasbg.fr